

```
In [1]: import pandas as pd
df = pd.read_csv(r'C:\Users\KHAN\Desktop\Pakistan e-commerce.csv')
```

C:\Users\KHAN\AppData\Local\Temp\ipykernel_2596\947329019.py:2: DtypeWarning: Columns (1,2,3,7,8,9,11,12,13,14,17,18,19) have mixed types. Specify dtype option on import or set low_memory=False.

```
df = pd.read_csv(r'C:\Users\KHAN\Desktop\pak.csv')
```

```
In [2]: print(df)
```

	item_id	status	created_at \
0	211131.0	complete	7/1/2016
1	211133.0	canceled	7/1/2016
2	211134.0	canceled	7/1/2016
3	211135.0	complete	7/1/2016
4	211136.0	order_refunded	7/1/2016
...	...	...	...
1048570	NaN	NaN	NaN
1048571	NaN	NaN	NaN
1048572	NaN	NaN	NaN
1048573	NaN	NaN	NaN
1048574	NaN	NaN	NaN

	sku	price \
0	kreations_YI 06-L	1950.0
1	kcc_Buy 2 Frey Air Freshener & Get 1 Kasual Bo...	240.0
2	Ego_UP0017-999-MR0	2450.0
3	kcc_krone deal	360.0
4	BK7010400AG	555.0
...	...	...
1048570	NaN	NaN
1048571	NaN	NaN
1048572	NaN	NaN
1048573	NaN	NaN
1048574	NaN	NaN

	qty_ordered	grand_total	increment_id	category_name_1 \
0	1.0	1950.0	100147443	Women's Fashion
1	1.0	240.0	100147444	Beauty & Grooming
2	1.0	2450.0	100147445	Women's Fashion
3	1.0	60.0	100147446	Beauty & Grooming
4	2.0	1110.0	100147447	Soghaat
...	...	...	...	...
1048570	NaN	NaN	NaN	NaN
1048571	NaN	NaN	NaN	NaN
1048572	NaN	NaN	NaN	NaN
1048573	NaN	NaN	NaN	NaN
1048574	NaN	NaN	NaN	NaN

	sales_commission_code	...	Month	Customer	Since	M-Y	FY	\
0	\N	...	7.0	2016-7	7-2016	FY17		
1	\N	...	7.0	2016-7	7-2016	FY17		

2		\N	...	7.0	2016-7	7-2016	FY17
3	R-FSD-52352		...	7.0	2016-7	7-2016	FY17
4		\N	...	7.0	2016-7	7-2016	FY17
...		...	...	...	...	...	...
1048570		NaN	...	NaN	NaN	NaN	NaN
1048571		NaN	...	NaN	NaN	NaN	NaN
1048572		NaN	...	NaN	NaN	NaN	NaN
1048573		NaN	...	NaN	NaN	NaN	NaN
1048574		NaN	...	NaN	NaN	NaN	NaN

	Customer ID	Unnamed: 21	Unnamed: 22	Unnamed: 23	Unnamed: 24	\
0	1.0	NaN	NaN	NaN	NaN	
1	2.0	NaN	NaN	NaN	NaN	
2	3.0	NaN	NaN	NaN	NaN	
3	4.0	NaN	NaN	NaN	NaN	
4	5.0	NaN	NaN	NaN	NaN	
...	...	...	...	...	...	
1048570	NaN	NaN	NaN	NaN	NaN	
1048571	NaN	NaN	NaN	NaN	NaN	
1048572	NaN	NaN	NaN	NaN	NaN	
1048573	NaN	NaN	NaN	NaN	NaN	
1048574	NaN	NaN	NaN	NaN	NaN	

	Unnamed: 25
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN
...	...
1048570	NaN
1048571	NaN
1048572	NaN
1048573	NaN
1048574	NaN

[1048575 rows x 26 columns]

```
In [3]: print(df.info())
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1048575 entries, 0 to 1048574
Data columns (total 26 columns):
#   Column                Non-Null Count  Dtype
---  -
0   item_id               584524 non-null  float64
1   status                584509 non-null  object
2   created_at            584524 non-null  object
3   sku                   584504 non-null  object
4   price                 584524 non-null  float64
5   qty_ordered           584524 non-null  float64
6   grand_total           584524 non-null  float64
7   increment_id          584524 non-null  object
8   category_name_1       584360 non-null  object
9   sales_commission_code 447346 non-null  object
10  discount_amount       584524 non-null  float64
11  payment_method        584524 non-null  object
12  Working Date          584524 non-null  object
13  BI Status             584524 non-null  object
14  MV                    584524 non-null  object
15  Year                  584524 non-null  float64
16  Month                 584524 non-null  float64
17  Customer Since        584513 non-null  object
18  M-Y                   584524 non-null  object
19  FY                    584524 non-null  object
20  Customer ID           584513 non-null  float64
21  Unnamed: 21           0 non-null       float64
22  Unnamed: 22           0 non-null       float64
23  Unnamed: 23           0 non-null       float64
24  Unnamed: 24           0 non-null       float64
25  Unnamed: 25           0 non-null       float64
dtypes: float64(13), object(13)
memory usage: 208.0+ MB
None

```

```
In [15]: print(df.duplicated().sum())
```

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```
In [17]: df = df[['status', 'created_at', 'sku', 'price', 'qty_ordered', 'grand_total', \
                'category_name_1', 'discount_amount', 'payment_method', 'Year', 'Month']]
```

```
df = df.drop_duplicates()  
print(df)
```

	status	created_at	\
0	complete	7/1/2016	
1	canceled	7/1/2016	
2	canceled	7/1/2016	
3	complete	7/1/2016	
4	order_refunded	7/1/2016	
...	...	...	
584520	processing	8/28/2018	
584521	processing	8/28/2018	
584522	processing	8/28/2018	
584523	processing	8/28/2018	
584524	processing	8/28/2018	

	sku	price	\
0	kreations_YI 06-L	1950.0	
1	kcc_Buy 2 Frey Air Freshener & Get 1 Kasual Bo...	240.0	
2	Ego_UP0017-999-MR0	2450.0	
3	kcc_krone deal	360.0	
4	BK7010400AG	555.0	
...	...	...	
584520	MATHUA5AF70A7D1E50A	35599.0	
584521	MATSAM5B6D7208C6D30	129999.0	
584522	MATSAM5B1509B4696EA	87300.0	
584523	MATSAM5B10F91A9B6AB	108640.0	
584524	MATSAM5B10F91A9B6AB	108640.0	

	qty_ordered	grand_total	category_name_1	discount_amount	\
0	1.0	1950.0	Women's Fashion	0.0	
1	1.0	240.0	Beauty & Grooming	0.0	
2	1.0	2450.0	Women's Fashion	0.0	
3	1.0	60.0	Beauty & Grooming	300.0	
4	2.0	1110.0	Soghaat	0.0	
...	...	...	...	...	
584520	1.0	35899.0	Mobiles & Tablets	0.0	
584521	2.0	652178.0	Mobiles & Tablets	0.0	
584522	2.0	652178.0	Mobiles & Tablets	0.0	
584523	2.0	652178.0	Mobiles & Tablets	0.0	
584524	2.0	652178.0	unknown	0.0	

	payment_method	Year	Month
0	cod	2016.0	7.0
1	cod	2016.0	7.0

```

2          cod  2016.0    7.0
3          cod  2016.0    7.0
4          cod  2016.0    7.0
...
584520  bankalfalah  2018.0    8.0
584521  bankalfalah  2018.0    8.0
584522  bankalfalah  2018.0    8.0
584523  bankalfalah  2018.0    8.0
584524  bankalfalah  2018.0    8.0

```

[439084 rows x 11 columns]

```
In [8]: print(df.isnull().sum())
```

```

status          464066
created_at      464051
sku             464071
price           464051
qty_ordered     464051
grand_total     464051
category_name_1 464215
discount_amount 464051
payment_method  464051
Year            464051
Month           464051
dtype: int64

```

```
In [9]: df['category_name_1'] = df['category_name_1'].fillna('unknown')
```

```
In [13]: df[['status', 'created_at', 'sku', 'price', 'qty_ordered', 'grand_total']] = \
          df[['status', 'created_at', 'sku', 'price', 'qty_ordered', 'grand_total']].ffill()
df[['discount_amount', 'payment_method', 'Year', 'Month']] = \
          df[['discount_amount', 'payment_method', 'Year', 'Month']].ffill()
print(df)
```

	status	created_at	\
0	complete	7/1/2016	
1	canceled	7/1/2016	
2	canceled	7/1/2016	
3	complete	7/1/2016	
4	order_refunded	7/1/2016	
...	...	...	
1048570	processing	8/28/2018	
1048571	processing	8/28/2018	
1048572	processing	8/28/2018	
1048573	processing	8/28/2018	
1048574	processing	8/28/2018	

	sku	price	\
0	kreations_YI 06-L	1950.0	
1	kcc_Buy 2 Frey Air Freshener & Get 1 Kasual Bo...	240.0	
2	Ego_UP0017-999-MR0	2450.0	
3	kcc_krone deal	360.0	
4	BK7010400AG	555.0	
...	...	...	
1048570	MATSAM5B10F91A9B6AB	108640.0	
1048571	MATSAM5B10F91A9B6AB	108640.0	
1048572	MATSAM5B10F91A9B6AB	108640.0	
1048573	MATSAM5B10F91A9B6AB	108640.0	
1048574	MATSAM5B10F91A9B6AB	108640.0	

	qty_ordered	grand_total	category_name_1	discount_amount	\
0	1.0	1950.0	Women's Fashion	0.0	
1	1.0	240.0	Beauty & Grooming	0.0	
2	1.0	2450.0	Women's Fashion	0.0	
3	1.0	60.0	Beauty & Grooming	300.0	
4	2.0	1110.0	Soghaat	0.0	
...	...	...	...	...	
1048570	2.0	652178.0	unknown	0.0	
1048571	2.0	652178.0	unknown	0.0	
1048572	2.0	652178.0	unknown	0.0	
1048573	2.0	652178.0	unknown	0.0	
1048574	2.0	652178.0	unknown	0.0	

	payment_method	Year	Month
0	cod	2016.0	7.0
1	cod	2016.0	7.0


```
2          cod  2016.0    7.0
3          cod  2016.0    7.0
4          cod  2016.0    7.0
...
1048570  bankalfalah  2018.0    8.0
1048571  bankalfalah  2018.0    8.0
1048572  bankalfalah  2018.0    8.0
1048573  bankalfalah  2018.0    8.0
1048574  bankalfalah  2018.0    8.0
```

[1048575 rows x 11 columns]

```
In [7]: df['Date'] = pd.to_datetime(df[['Year', 'Month']].assign(day=1))
        print(df)
```

	status	created_at	\
0	complete	7/1/2016	
1	canceled	7/1/2016	
2	canceled	7/1/2016	
3	complete	7/1/2016	
4	order_refunded	7/1/2016	
...	...	...	
584520	processing	8/28/2018	
584521	processing	8/28/2018	
584522	processing	8/28/2018	
584523	processing	8/28/2018	
584524	processing	8/28/2018	

	sku	price	\
0	kreations_YI 06-L	1950.0	
1	kcc_Buy 2 Frey Air Freshener & Get 1 Kasual Bo...	240.0	
2	Ego_UP0017-999-MR0	2450.0	
3	kcc_krone deal	360.0	
4	BK7010400AG	555.0	
...	...	...	
584520	MATHUA5AF70A7D1E50A	35599.0	
584521	MATSAM5B6D7208C6D30	129999.0	
584522	MATSAM5B1509B4696EA	87300.0	
584523	MATSAM5B10F91A9B6AB	108640.0	
584524	MATSAM5B10F91A9B6AB	108640.0	

	qty_ordered	grand_total	category_name_1	discount_amount	\
0	1.0	1950.0	Women's Fashion	0.0	
1	1.0	240.0	Beauty & Grooming	0.0	
2	1.0	2450.0	Women's Fashion	0.0	
3	1.0	60.0	Beauty & Grooming	300.0	
4	2.0	1110.0	Soghaat	0.0	
...	...	...	...	...	
584520	1.0	35899.0	Mobiles & Tablets	0.0	
584521	2.0	652178.0	Mobiles & Tablets	0.0	
584522	2.0	652178.0	Mobiles & Tablets	0.0	
584523	2.0	652178.0	Mobiles & Tablets	0.0	
584524	2.0	652178.0	unknown	0.0	

	payment_method	Year	Month	Date
0	cod	2016.0	7.0	2016-07-01
1	cod	2016.0	7.0	2016-07-01

```
2          cod  2016.0    7.0 2016-07-01
3          cod  2016.0    7.0 2016-07-01
4          cod  2016.0    7.0 2016-07-01
...
584520  bankalfalah  2018.0    8.0 2018-08-01
584521  bankalfalah  2018.0    8.0 2018-08-01
584522  bankalfalah  2018.0    8.0 2018-08-01
584523  bankalfalah  2018.0    8.0 2018-08-01
584524  bankalfalah  2018.0    8.0 2018-08-01
```

```
[584525 rows x 12 columns]
```

```
In [11]: del df['sku']
         print(df)
```

	status	created_at	price	qty_ordered	grand_total	\
0	complete	7/1/2016	1950.0	1.0	1950.0	
1	canceled	7/1/2016	240.0	1.0	240.0	
2	canceled	7/1/2016	2450.0	1.0	2450.0	
3	complete	7/1/2016	360.0	1.0	60.0	
4	order_refunded	7/1/2016	555.0	2.0	1110.0	
...	...	...	...	...	...	
584520	processing	8/28/2018	35599.0	1.0	35899.0	
584521	processing	8/28/2018	129999.0	2.0	652178.0	
584522	processing	8/28/2018	87300.0	2.0	652178.0	
584523	processing	8/28/2018	108640.0	2.0	652178.0	
584524	processing	8/28/2018	108640.0	2.0	652178.0	

	category_name_1	discount_amount	payment_method	Year	Month	\
0	Women's Fashion	0.0	cod	2016.0	7.0	
1	Beauty & Grooming	0.0	cod	2016.0	7.0	
2	Women's Fashion	0.0	cod	2016.0	7.0	
3	Beauty & Grooming	300.0	cod	2016.0	7.0	
4	Soghaat	0.0	cod	2016.0	7.0	
...	...	...	...	...	...	
584520	Mobiles & Tablets	0.0	bankalfalah	2018.0	8.0	
584521	Mobiles & Tablets	0.0	bankalfalah	2018.0	8.0	
584522	Mobiles & Tablets	0.0	bankalfalah	2018.0	8.0	
584523	Mobiles & Tablets	0.0	bankalfalah	2018.0	8.0	
584524	unknown	0.0	bankalfalah	2018.0	8.0	

	Date
0	2016-07-01
1	2016-07-01
2	2016-07-01
3	2016-07-01
4	2016-07-01
...	...
584520	2018-08-01
584521	2018-08-01
584522	2018-08-01
584523	2018-08-01
584524	2018-08-01

[584525 rows x 11 columns]

```
In [14]: df['category_name_1'] = df['category_name_1'].str.replace(['unknown', '\\N'], 'Others')
```

```
In [3]: print(df['category_name_1'].unique())
print(df['status'].unique())
print(df['payment_method'].unique())
```

```
["Women'S Fashion" 'Beauty & Grooming' 'Soghaat' 'Mobiles & Tablets'
 'Appliances' 'Home & Living' "Men'S Fashion" 'Kids & Baby' 'Others'
 'Entertainment' 'Computing' 'Superstore' 'Health & Sports' 'Books'
 'School & Education']
['complete' 'canceled' 'order_refunded' 'received' 'refund' 'closed'
 'fraud' 'holded' 'exchange' 'pending_paypal' 'paid' 'Others' 'cod'
 'pending' 'processing' 'payment_review']
['cod' 'ublccreditcard' 'mygateway' 'customercredit' 'cashatdoorstep'
 'mcblite' 'internetbanking' 'marketingexpense' 'productcredit'
 'financesettlement' 'Payaxis' 'jazzvoucher' 'jazzwallet' 'Easypay'
 'Easypay_MA' 'easypay_voucher' 'bankalfalah' 'apg']
```

```
In [9]: df['category_name_1'] = df['category_name_1'].str.replace("Women'S Fashion", "Women's Fashion")\
        .str.replace("Men'S Fashion", "Men's Fashion").str.strip()

df['status'] = df['status'].str.replace('cod', 'COD').str.title().str.strip()
df['status'] = np.where(df['status'].str.contains\
('Paypal|Pending|Review|Holded|Processing|Paid|Cod', na=False), 'Processing/Active', df['status'])
df['status'] = np.where(df['status'].str.contains\
('Completed|Delivered', na=False), 'Completed', df['status'])
df['status'] = np.where(df['status'].str.contains\
('Canceled|Refunded|Refund|Closed|Exchange', na=False), 'Canceled/Refunded', df['status'])
df['status'] = np.where(df['status'].str.contains\
('Fraud|Others', na=False), 'Problematic Orders', df['status'])

df['payment_method'] = df['payment_method'].str.title().str.strip()
df['payment_method'] = np.where(df['payment_method'].str.contains\
('Cod|Cashatdoorstep', na=False), 'Cash Payment', df['payment_method'])
df['payment_method'] = np.where(df['payment_method'].str.contains\('Ublcreditcard|\
Customercredit|Productcredit|Mcblite', na=False), 'Credit Card', df['payment_method'])
df['payment_method'] = np.where(df['payment_method'].str.contains('Mygateway|Internetbanking\
|Bankalfalah|Apg|Financesettlement', na=False), 'Bank Transfers', df['payment_method'])
df['payment_method'] = np.where(df['payment_method'].str.contains\
('Easypay|Easy_Ma|Voucher', na=False), 'Easypaisa Variants', df['payment_method'])
```

```
df['payment_method'] = np.where(df['payment_method'].str.contains('Jazzvoucher|\nJazzwallet|Payaxis|Marketingexpense', na=False), 'Jazz & Other Wallets', df['payment_method'])
```

```
In [4]: import numpy as np
```

```
In [10]: print(df['category_name_1'].unique())
print(df['status'].unique())
print(df['payment_method'].unique())
```

```
["Women's Fashion" 'Beauty & Grooming' 'Soghaat' 'Mobiles & Tablets'
 'Appliances' 'Home & Living' "Men's Fashion" 'Kids & Baby' 'Others'
 'Entertainment' 'Computing' 'Superstore' 'Health & Sports' 'Books'
 'School & Education']
['Complete' 'Canceled/Refunded' 'Received' 'Problematic Orders'
 'Processing/Active']
['Cash Payment' 'Credit Card' 'Bank Transfers' 'Jazz & Other Wallets'
 'Easypaisa Variants']
```

```
In [11]: df[['grand_total', 'Year', 'Month', 'discount_amount', 'qty_ordered', 'price']] = \
df[['grand_total', 'Year', 'Month', 'discount_amount', 'qty_ordered', 'price']].astype(int)
```

```
In [12]: print(df)
```

	Unnamed: 0	status	created_at	price	qty_ordered	\
0	0	Complete	7/1/2016	1950	1	
1	1	Canceled/Refunded	7/1/2016	240	1	
2	2	Canceled/Refunded	7/1/2016	2450	1	
3	3	Complete	7/1/2016	360	1	
4	4	Canceled/Refunded	7/1/2016	555	2	
...	...	...	...	...	...	
584520	584520	Processing/Active	8/28/2018	35599	1	
584521	584521	Processing/Active	8/28/2018	129999	2	
584522	584522	Processing/Active	8/28/2018	87300	2	
584523	584523	Processing/Active	8/28/2018	108640	2	
584524	584524	Processing/Active	8/28/2018	108640	2	

	grand_total	category_name_1	discount_amount	payment_method	Year	\
0	1950	Women's Fashion	0	Cash Payment	2016	
1	240	Beauty & Grooming	0	Cash Payment	2016	
2	2450	Women's Fashion	0	Cash Payment	2016	
3	60	Beauty & Grooming	300	Cash Payment	2016	
4	1110	Soghaat	0	Cash Payment	2016	
...	...	...	...	...	...	
584520	35899	Mobiles & Tablets	0	Bank Transfers	2018	
584521	652178	Mobiles & Tablets	0	Bank Transfers	2018	
584522	652178	Mobiles & Tablets	0	Bank Transfers	2018	
584523	652178	Mobiles & Tablets	0	Bank Transfers	2018	
584524	652178	Others	0	Bank Transfers	2018	

	Month	Date
0	7	2016-07-01
1	7	2016-07-01
2	7	2016-07-01
3	7	2016-07-01
4	7	2016-07-01
...	...	...
584520	8	2018-08-01
584521	8	2018-08-01
584522	8	2018-08-01
584523	8	2018-08-01
584524	8	2018-08-01

[584525 rows x 12 columns]

```
In [16]: import os
desktop = os.path.join(os.path.expanduser('~'), 'Desktop')
df.to_csv(r'C:\Users\KHAN\Desktop\Pakistan e-commerce.csv', index=False)
```